

Pairing symmetry and superconductivity in $\text{La}_3\text{Ni}_2\text{O}_7$ thin films

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The recent discovery of superconductivity with a transition temperature T_c over 40 K in $\text{La}_3\text{Ni}_2\text{O}_7$ and $(\text{La},\text{Pr})_3\text{Ni}_2\text{O}_7$ thin films at ambient pressure marks an important step in the field of nickelate superconductors. Here, we perform a renormalized mean-field theory study of the superconductivity in $\text{La}_3\text{Ni}_2\text{O}_7$ thin films, using a bilayer two-orbital $t - J$ model. Our result reveals an $s_{\pm}$ -wave pairing symmetry driven by the strong interlayer superexchange coupling of d_{z^2} orbital, resembling the pressurized bulk case. Also, we roughly reproduce the experimentally observed nodeless shape of the superconducting gap at the β pocket and the superconducting T_c . In addition, by analysing the orbital-resolved pairing configurations and their projections onto Fermi surface, we find that the nodeless feature of β pocket is related to the interlayer pairing within both d_{z^2} and $d_{x^2-y^2}$ orbitals. Moreover, we identify a formation of the inplane inter-orbital d -wave pairing between d_{z^2} and $d_{x^2-y^2}$ orbitals, which can even enhance the dominated interlayer $s_{\pm}$ -wave. Our study particularly highlights the diverse relations of different pairing channels in $\text{La}_3\text{Ni}_2\text{O}_7$ that holds a complex Fermi surface.

I. INTRODUCTION

The discovery of superconductivity with a transition temperature T_c near 80 K in the Ruddlesden-Popper (RP) bilayer nickelate $\text{La}_3\text{Ni}_2\text{O}_7$ (LNO) [1] has gained widespread interests in the field of high- T_c superconductivity, and it was soon followed by another observation of superconductivity in trilayer nickelate $\text{La}_4\text{Ni}_3\text{O}_{10}$ with $T_c=20\text{-}30$ K [2, 3], both under high pressure. The discovery of RP nickelate-based superconductors has marked a breakthrough in the field of high- T_c superconductivity, and therefore raised considerable subsequent investigations both theoretically [4–38] and experimentally [3, 39–45]. However, the need for high pressure has largely confined some critical experimental techniques for a direct probe of the superconducting phase, such as angle resolved photoemission spectroscopy (ARPES) and scanning tunneling microscopy (STM), which prevents a decisive confirmation of the superconducting pairing among various theoretical proposals [16, 34]. On the other hand, achieving ambient-pressure superconductivity is a crucial step for practical considerations, and in this respect, there are some suggestions on the possible route, such as using chemical substitute [46–48] or applying strain [49–52].

Very recently, superconductivity with T_c over 40 K in LNO [54] and $(\text{La},\text{Pr})_3\text{Ni}_2\text{O}_7$ [55] thin films at ambient pressure have been reported. The X-ray absorption spectroscopy (XAS) analysis and scanning transmission electron microscopy (STEM) reveal that, the Ni ions in LNO thin films maintain a mixed valence state and the apical Ni-O-Ni bond angle approaches 180° [54], which

closely resembles the high-pressure bulk phase of LNO. The ARPES measurement [56] in thin films also confirms the dominant Ni- $d_{x^2-y^2}$ and d_{z^2} orbital spectral weights around the Fermi energy, which is consistent with DFT calculations [51–53, 57, 58]. Moreover, a direct ARPES measurement of the superconducting electronic structure on the Fermi surface (FS) was performed recently [59], which reveals a notable gap opening on the β FS sheet and without showing any node along the Brillouin zone (BZ) diagonal direction. This observation is accompanied by another STM measurement on superconducting state [60], which shows a two-gap structure on the FS and a fitting to such gaps demonstrates a preferable anisotropic s -wave pairing. The two results are generally prone to the idea of the predominant $s_{\pm}$ -wave pairing

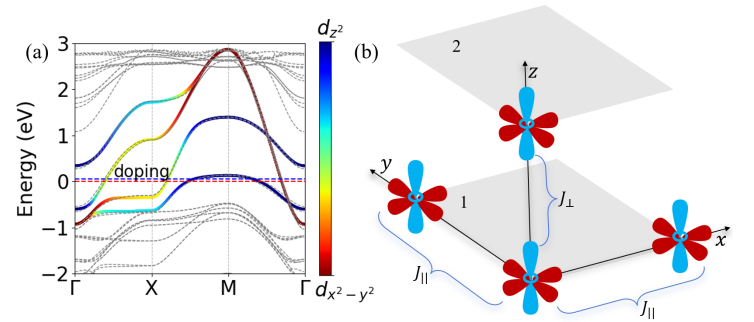


FIG. 1. (a) Band structure of the 4-orbital tight-binding model adopted from the DFT simulation of the double-stacked $\text{La}_3\text{Ni}_2\text{O}_7$ thin film [53]. The red dashed line denotes the undoped case from DFT while the blue one denotes the slightly electron doped case as is used in our study. (b) Schematic of $d_{x^2-y^2}$ (red) and d_{z^2} (blue) orbitals of the model. Two major superexchange couplings $J_{\perp}$, $J_{\parallel}$ are drawn. See also the main text.

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symmetry in the system as predicted by several theoretical studies [7, 16, 18, 35, 57].

In this paper, using a bilayer $t - J$ model that captures the key ingredient of LNO thin films, we perform a renormalized mean-field theory (RMFT) [7] calculation of the superconductivity. Our result reveals the $s_{\pm}$ -wave pairing and reproduces the experimentally observed nodeless shape of the superconducting gap at β pocket in Ref. [59]. To gain insight, we also analyse the orbital-resolved pairing bonds and their projections onto Fermi surface. Our study highlights the diverse relations of different pairing channels in LNO that holds a complex Fermi surface.

II. MODEL AND METHOD

The bilayer two-orbital $t - J$ model is defined as

$$\mathcal{H} = \mathcal{H}_t + \mathcal{H}_J, \quad (1)$$

$$\mathcal{H}_t = \sum_{ij} \sum_{\mu\nu} \sum_{\sigma} t_{ij}^{\mu\nu} c_{i\mu\sigma}^{\dagger} c_{j\nu\sigma} - \mu \sum_{i\mu\sigma} n_{i\mu\sigma} \quad (2)$$

$$\begin{aligned} \mathcal{H}_J = & J_{\perp} \sum_i \mathbf{S}_{iz_1} \cdot \mathbf{S}_{iz_2} + J_{\parallel} \sum_{\langle ij \rangle}^{\mu=x_1, x_2} \mathbf{S}_{i\mu} \cdot \mathbf{S}_{j\mu} \\ & + J_{xz} \sum_{\langle ij \rangle}^{\mu\nu=x_1 z_1, x_2 z_2} \mathbf{S}_{i\mu} \cdot \mathbf{S}_{j\nu} - J_H \sum_i^{\mu\nu=x_1 z_1, x_2 z_2} \mathbf{S}_{i\mu} \cdot \mathbf{S}_{i\nu}. \end{aligned} \quad (3)$$

In this model, the basis is $\Psi = (d_{z_1}, d_{x_1}, d_{z_2}, d_{x_2})^T$ that contains d_{z_2} and $d_{x^2-y^2}$ orbitals for the first and second layers, as shown in Fig. 1(b). $\mathcal{H}_t$ is the tight-binding Hamiltonian adopted from DFT simulation for the double-stacked LNO thin films [53]. Here, $ij, \mu\nu, \sigma$ denote the lattice, orbital, and spin indices, respectively. The band structure is shown in Fig. 1(a), in which a small electron doping level with chemical potential $\mu = 0.05$ is used in our calculation to better agree with the experiments, which gives the electron filling $n_z=0.8$, $n_x=0.58$. For the Heisenberg term $\mathcal{H}_J$, $J_{\perp}$ is the inter-layer superexchange coupling of d_{z_2} orbitals, and $J_{\parallel}, J_{xz}$ are the inplane nearest neighbor superexchange coupling of $d_{x^2-y^2}$ orbital, and superexchange coupling between $d_{x^2-y^2}$ and d_{z_2} , respectively. Also, the Hund's coupling J_H is included for generality. To quantitatively estimate $J_{\perp}, J_{\parallel}, J_{xz}$, we perform the exact diagonalization (ED) calculation and obtain $J_{\perp}=0.135$ eV, $J_{\parallel}=0.084$ eV, $J_{xz}=0.03$ eV. Taking $J_{\perp}$ as an example, ED is performed for a 5-site chain containing two d_{z_2} and three p_z orbitals along the z axis, with $p - d$ hoppings obtained from DFT [53]. Under the atomic subspace that corresponds to half-filling of d_{z_2} orbital ($N=8, S_z=0$), $J_{\perp}$ is determined as the energy difference of the lowest singlet and triplet states [8]. Similar method is applied in Refs. [61, 62]. Besides, a realistic $J_H=1$ eV is also used in this study.

The above $t - J$ model is solved under the RMFT. The RMFT [63] follows a conventional mean-field decomposition of spin exchange, which yields the order parameters

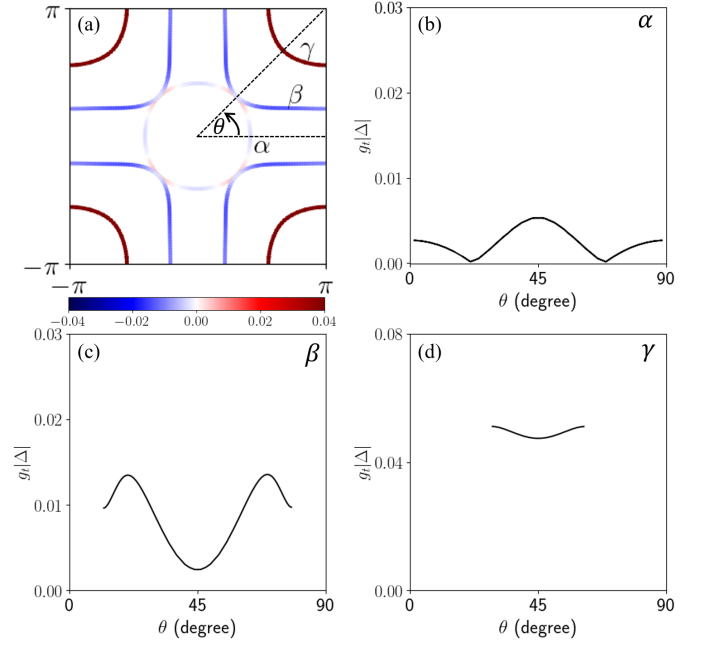


FIG. 2. (a) Superconducting gap $g_t \Delta$ projected onto FS for $\text{La}_3\text{Ni}_2\text{O}_7$ thin films. (b-d) $g_t |\Delta|$ as a function of angle θ for α , β and γ pockets, respectively. θ is indicated in (a).

$\chi_{ij}^{\mu\nu} = \langle c_{i\mu\uparrow}^{\dagger} c_{j\nu\uparrow} \rangle$ and $\Delta_{ij}^{\mu\nu} = \langle c_{i\mu\uparrow}^{\dagger} c_{j\nu\downarrow}^{\dagger} \rangle$ under the spin symmetry. Besides, $\mathcal{H}_t$ and $\mathcal{H}_J$ are respectively coupled by the Gutzwiller renormalization factors g_t and g_J to recover the strong renormalization from correlation [64]. The obtained quadratic Hamiltonian is then calculated self-consistently, during which the electron densities are fixed to certain level. The eventual pairing symmetry can be determined from the phase structure of various pairing bonds $\Delta_{ij}^{\mu\nu}$. For the inplane pairing, this is done by allowing Δ along x and y directions to evolve independently, so a final $s_{\pm}, d$ wave or $s + id$ wave is possible depending on the superexchanges. We refer reader to Ref. [7] for details of the method.

III. PAIRING SYMMETRY AND SUPERCONDUCTING GAP

Figure 2(a) presents the superconducting gap matrix $[g_t \Delta]_{\mu\nu}$ projected onto FS, which demonstrates that β, γ pockets are nodeless and of opposite sign, while α pocket shows a sign flip. To better illustrate the gap features, Fig. 2(b-d) show the absolute value $g_t |\Delta|$ as a function of angle θ for α, β and γ pockets, respectively. For α pocket in Fig. 2(b), the gap demonstrates a node between $\theta = 45^\circ \sim 90^\circ$. For β pocket in Fig. 2(c), the gap shows a parabolic shape centering at $\theta = 45^\circ$, which gets slightly decrease when approaching the corner. Note that the gap shape as well as its magnitude around the BZ diagonal direction from our result are generally coordinated with the ARPES result [59], which suggests

that our RMFT captures the key ingredient in the LNO thin films. For γ pocket in Fig. 2(d), we find the gap magnitude is about 5 times larger than that in Fig. 2(b), with its minimum appearing at the BZ diagonal direction. Apparently, such large gap opening at γ pocket should be attributed to the strong interlayer superexchange coupling $J_{\perp}$, which is also consistent with previous calculations for bulk LNO [7, 16, 18, 25]. Overall, the pairing structure in Fig. 2 should be described as the $s_{\pm}$ -wave, which can also be seen from our result that besides the formation of the large interlayer pairing bond for d_{z^2} orbital ($\Delta_{\perp}^z$), the inplane pairing bonds have the same magnitude when along x, y directions for both $d_{x^2-y^2}$ and d_{z^2} orbitals ($\Delta_{\parallel}^x, \Delta_{\parallel}^z$). Note that, in our $t-J$ model only the pairing bonds $\Delta_{\perp}^z, \Delta_{\parallel}^x, \Delta_{\parallel}^z$ are associated with the applied $J_{\perp}, J_{\parallel}, J_{xz}$, respectively, while other terms like the interlayer pairing of $d_{x^2-y^2}$ orbital ($\Delta_{\perp}^x$) do not have the corresponding J and should not appear in the calculation. However, they can still gain finite values as $\Delta_{ij}^{\mu\nu} = \langle nk | c_{i\mu}^{\dagger} c_{j\nu}^{\dagger} | nk \rangle$, which reflects the pairing tendency towards a global coherent state at the mean-field level. In the following we will dive into the role of different pairing bonds to see how they interplay and collectively give rise to the result in Fig. 2.

IV. ORBITAL-RESOLVED SUPERCONDUCTING GAP

Figure. 3 presents the orbital-resolved superconducting gap for (a) $\Delta_{\perp}^z$, (b) $\Delta_{\parallel}^z$, (c) $\Delta_{\perp}^x$, and (d) $\Delta_{\parallel}^x$, respectively. For $\Delta_{\perp}^z$ in Fig. 3(a), the gap is positive for the bonding branch of α, γ pockets, and is negative for the anti-bonding branch of β pocket. For $\Delta_{\parallel}^z$ in Fig. 3(b), the gap structure clearly demonstrates the sign-flip $s_{\pm}$ -wave with the node appearing at the extended BZ boundary (diamond shape). Also, another node at the BZ diagonal direction in both Fig. 3(a,b) is identified due to the vanishing of hybridization. For $\Delta_{\perp}^x, \Delta_{\parallel}^x$ in Fig. 3(c,d), similar behavior is observed as compared to Fig. 3(a,b), but with a much smaller magnitude. It is worth noting that the phases in Fig. 3(a-d) all can be understood in terms of maximizing the total gap magnitude. Because the large $J_{\perp}$ will guarantees $\Delta_{\perp}^z$ to be the leading pairing, other pairings in Fig. 3(c-d) will have to fit into it. First, Fig. 3(c) will choose the same sign that matches Fig. 3(a) ($\text{sgn}\Delta_{\perp}^z = \text{sgn}\Delta_{\perp}^x$). This does not conflict with their spatial configuration as the two sides are both connected by J_H . This result clearly explains the experimentally observed nodeless shape of β pocket as a joint effort of interlayer pairing for both $d_{x^2-y^2}$ and d_{z^2} orbitals. Second, due to the inplane $\cos k_x + \cos k_y$ form factor in Fig. 3(b,d), cancellation of the gap in certain region of FS is unavoidable, which inherently reflects the “pairing frustration”. In this case $\Delta_{\parallel}^z$ will choose to maximize the gap for γ pocket at the price of minimizing the gap for α pocket, while $\Delta_{\parallel}^x$ will no doubt choose to maximize the

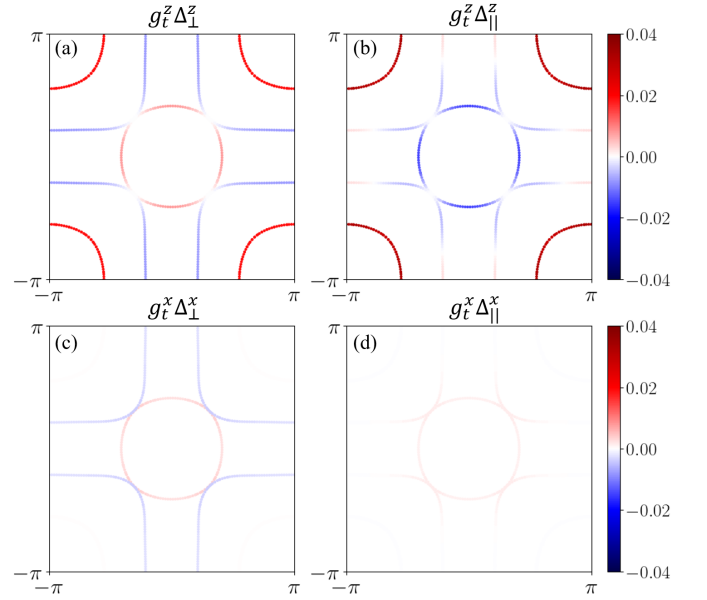


FIG. 3. Orbital-resolved superconducting energy gap on FS for (a) $\Delta_{\perp}^z$, (b) $\Delta_{\parallel}^z$, (c) $\Delta_{\perp}^x$, and (d) $\Delta_{\parallel}^x$. $g_t^{z/x}$ is the renormalization factor for $d_{z^2}/d_{x^2-y^2}$ orbital. $\Delta_{\perp/\parallel}^{z/x}$ denotes the interlayer/inplane pairing bond for $d_{z^2}/d_{x^2-y^2}$ orbital.

gap for α pocket, with slight cancelation at β pocket. In this sense, “pairing frustration” is more likely to occur in system with multi-orbital feature and complex Fermi surface where there can not find a single pairing symmetry to perfectly fit in with. More insights into it and the “gap maximization” via free energy functional is provided in the Appendix. In LNO, the obtained gap relation is $\text{sgn}\Delta_{\perp}^z = \text{sgn}\Delta_{\perp}^x = \text{sgn}\Delta_{\parallel}^x = -\text{sgn}\Delta_{\parallel}^z$. General agreements can be found in Refs. [18, 31]. Finally, concerning the inplane inter-orbital pairing $\Delta_{\parallel}^{xz}$, surprisingly, we find it actually forms a d -wave and can even enhance the dominated $\Delta_{\perp}^z$. This is a unique feature for LNO tetrahedral lattice where the two distinct pairings can coexist within A_{1g} channel, see Appendix and Ref. [65] for details.

V. TRANSITION TEMPERATURE

In Fig. 4, we show the superconducting gap as a function of temperature T , which reveals two dominated pairing bonds from d_{z^2} orbitals (red), in line with Fig. 3. As increasing T , these gaps decrease in a mean-field manner and simultaneously drop to zero at $T_c \approx 60$ K. Such value is comparable to the experimental reported $T_c = 40 \sim 60$ K in the thin films [49, 54, 55]. Given that our previous RMFT study also obtains an experimental comparable $T_c \approx 80$ K for the pressurized bulk LNO [7], it is reasonable to believe that the decrease of T_c in the thin films should originate from the decrease of $J_{\perp}$. This speculation draws us to the difference in the electronic structure, especially in the mainly concerned interlayer

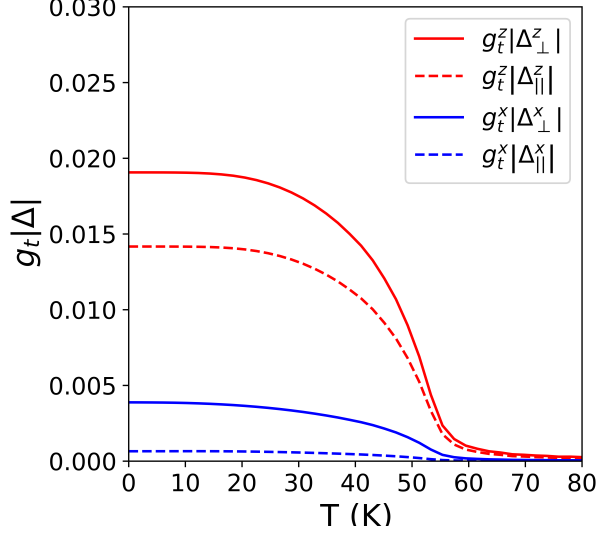


FIG. 4. Superconducting gap as a function of temperature for different pairing bonds. $g_t^{z/x}$ is the renormalization factor for $d_{z^2}/d_{x^2-y^2}$ orbital, and $\Delta_{\perp/\parallel}^{z/x}$ is the interlayer/inplane pairing bond for $d_{z^2}/d_{x^2-y^2}$ orbital.

hopping $t_\perp^z$ for d_{z^2} orbital. Notably, we find that the $J_\perp=0.135$ eV in this study is $\sim 25\%$ smaller than that of $J_\perp=0.18$ eV for the pressurized bulk [8] as estimated from ED, which is coordinated with a simple estimation using $J_\perp \propto (t_\perp^z)^2$ [4, 53]. Recently, a strain-tuning experiment in LNO thin films reports an increase of T_c from 10 K in the tensile-strained SrTiO₃ substrate to 60 K in the compressively strained LaAlO₃ substrate under 20 GPa of pressure [49]. This indicates a positive dependence of T_c on the lattice ratio c/a and, at first glance, violates the above understanding. However, we tend to believe $J_\perp$ can only determine the upper bound of T_c , while other factors like hybridization, density wave, and interlayer Josephson coupling can also have intricate impact, as also pointed out in Ref. [49, 66–71].

VI. DISCUSSION AND CONCLUSION

Our RMFT study of the superconductivity in LNO thin films has predicted an $s_\pm$ -wave pairing symmetry. The result largely resembles that of the pressurized bulk LNO [7, 16, 18, 25] giving the similarity in the electronic structure. Mathematically, our application of Gutzwiller projection [18, 63] on a multi-orbital $t-J$ model imposes a constraint on the renormalization condition to make the renormalization factor depends only on the orbital densities [7, 72], so the RMFT formation can be solved self-consistently like the standard Bardeen-Cooper-Schrieffer (BCS) theory. Therefore, it is expected that the obtained $s_\pm$ -wave mainly reflects pairing preference for the LNO FS topology in the presence of $J_\perp$ [73]. Nevertheless,

we find that the $s_\pm$ -wave is quite robust under various approaches from weak to strong couplings, which convinces us that the fluctuation of correlated atomic states might not significantly alter such scenario. In this scenario, the γ pocket plays a central role. Experimentally, there are two relevant ARPES measurements in which the γ pocket shows visible profile for that performed in the normal state [56] but invisible for that in the superconducting state [50]. From our result, this contradiction might be due to the opening of a large gap that exceeds the detected energy range, as shown Fig. 2(d). More experimental evidences are expected on this issue. Regarding the role of J_H , no significant impact is observed from our approach at the singlet pairing channel. We see that J_H mainly contributes to an effective interlayer correlation of $d_{x^2-y^2}$ orbitals.

To summarize, we perform a RMFT study of the superconductivity in the LNO thin films using a bilayer $t-J$ model. Our result reveals an $s_\pm$ -wave pairing symmetry driven by the interlayer superexchange coupling, resembling the bulk LNO case. Our result also roughly reproduces the experimentally reported nodeless shape of the superconducting gap at β pocket [59] as well as the superconducting T_c . In addition, we systematically analyse the relation between orbital-resolved pairing bonds and the gap structure on FS. The result shows that the nodeless gap of β pocket is related to the interlayer pairing within both d_{z^2} and $d_{x^2-y^2}$ orbitals. We also find the formation of inplane inter-orbital d -wave pairing between $d_{x^2-y^2}$ and d_{z^2} orbitals that can enhance the dominated interlayer $s_\pm$ -wave. The coexistence is robust symmetrically in La₃Ni₂O₇ lattice.

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Appendix A: Details on the pairing configurations for the bilayer $t-J$ model

In this section, we illustrate the role of different superconducting pairing bonds in our bilayer $t-J$ model in La₃Ni₂O₇ thin films. The bilayer two-orbital model in our main text is defined under the basis $\Psi_k = (d_{z_1}, d_{x_1}, d_{z_2}, d_{x_2})^T$, and the extension to the Nambu ba-

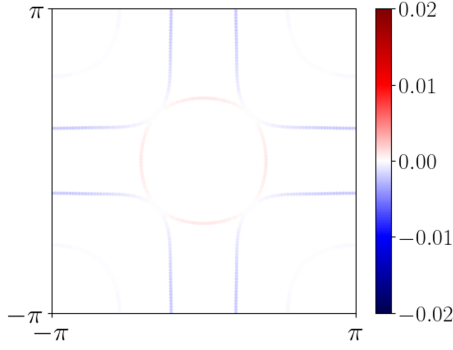


FIG. 6. Superconducting gap $g_t^{xz} \Delta_{||}^{xz}$ projected onto FS.

consistent approach, with no constraint on the symmetry, corresponding to a full optimization of F with respect to all relevant Δ_i , we find the relation of different pairing channels naturally manifests as the sign structure of the gaps. In case of $\text{La}_3\text{Ni}_2\text{O}_7$, it is due to the large $J_\perp$ that F will priority gain tremendous energy by forming $\Delta_\perp^z$, after that other Δ_i will form on top of that to see if it can further decrease F . This naturally explains the outcome of Fig. 2 in the main text.

To proceed with, we perform a transform to the bonding-antibonding representation as

$$H_\pm = H_{||} \pm H_\perp, \quad \Delta_\pm = \Delta_{||} \pm \Delta_\perp. \quad (\text{A8})$$

It is clear from this representation that, when along the Brillouin zone diagonal direction, the gaps for α, β, γ pockets respectively acquire the form

$$2\Delta_{||}^x(\cos k_x + \cos k_y) + \Delta_\perp^x, \quad (\text{A9})$$

$$2\Delta_{||}^x(\cos k_x + \cos k_y) - \Delta_\perp^x, \quad (\text{A10})$$

$$-2\Delta_{||}^z(\cos k_x + \cos k_y) + \Delta_\perp^z. \quad (\text{A11})$$

Eq. (A11) shows that, in order to maximize the gap at γ pocket around $\text{M}=(\pi, \pi)$ point, $-\Delta_{||}^z, \Delta_\perp^z$ have to be in antiphase. Such antiphase is also identified in Ref. [31], but there only $d_{x^2-y^2}$ orbital is relevant while d_{z^2} orbital has been integrated out a priori. But for $d_{x^2-y^2}$ orbital in our case, as demonstrated in Eq. (A9), they actually hold the same sign in order to maximize the gap at α pocket. For β pocket in Eq. (A10), a competition between $\Delta_{||}^x, \Delta_\perp^x$ occurs, and our result as well as the experimental evidence suggest that $4\Delta_{||}^x \cos k_x < \Delta_\perp^x$. We also note that in Ref. [18], a negative sign of $\Delta_{||}^x$ is revealed using the functional renormalization group calculation for a multiorbital atomic Coulomb interaction. Since the pairing configuration in Fig. 5 mainly reflects the pairing preference for the $\text{La}_3\text{Ni}_2\text{O}_7$ FS in the presence of $J_\perp$, the slight discrepancy may be attributed to other high-order processes besides $J_\perp$.

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